CRYPTOGRAPHY AND NETWORK SECURITY

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1. Caesar cipher - Fixed shift Encryption :

Given message : MEET AT NOON

Shift value : 4 position

(a) Reconstruct the encrypted message.

plaintext M E E T A T N O O N

Ciphertext Q I I X E X R S S R

Encrypted message : QIIXEXRSSR

(b) How does authorized member retrieve the original message ?

* The authorized member decrypts the message by shifting each letter four position backward in the alphabet.

* After applying the reverse shift and keeping the spaces in their original position, the plain text became plaintext : MEET AT NOON.

Conclusion : The caesar cipher is a simple substitution cipher in which each letter is replaced by another letter at a fixed distance in the alphabet.

2. Atbash cipher - Intercepted suspicious Message :

Given ciphertext : GSRH RH Z HVXIVG

(a) Deduce the most likely original message

* The ciphertext uses the atbash cipher, a substitution cipher that replace each letter with its opposite letter in the alphabets.

The alphabet mapping is :

Original : A B C D E F G H I J K L ...

Atbash : Z Y X W V U T S R Q P O N ...

*the reverse mapping applies to the second half of the alphabet.

Decryption :

* GSRH → THIS
* RH → IS
* Z → A
* HVXING → SECRET

Original message : THIS IS A SECRET

(b) Justify the answer :

* Reversed alphabet : Atbash maps A to Z, B to Y, C to X and so on.
* Word pattern : The two letter word RH decapts to I
* Meaningful sentence : The complete decapted message forms "THIS IS A SECRET"

: Atbash uses a fixed, one-to-one substitution without requires a secret key.

Conclusion :

* The Atbash cipher is easy to identify and encrypt and decrypt because its substitution rule is fixed and does not depend on a secret key

3. Vigenere cipher - Keyword based encryption.

Given plain text : HELLO

Keyword : KEY

* This vigenere cipher is a polyalphabetic substitution cipher that uses a repeating keyword to encrypt a message.

(a) Demonstrate the encryption process step by step:

Step 1 : Repeat the keyword.

* The keyword is repeated until it matches the length of a plaintext.

plaintext	H	E	L	L	O
keyword	K	E	Y	K	E

Step 2 : Convert letters into numerical values.

Use A=0, B=1, C=2, ... Z=25.

Plaintext	Value	keyletter	value
H	7	K	10
E	4	E	4
L	11	Y	24
L	11	K	10
O	14	E	4

Step-3: Apply encryption formula.

$$C = (P+K) \bmod 26$$

* H + K = 7 + 10 → 17 → R
* E + E = ~~10+10~~ 4+4 = 8 → I

Encrypted message → RIJVS

(b) Explain how the repeating keyword influences each character

* Each plaintext letter is combined with the corresponding keyword letter.
* When the keyword ends, it repeats from the beginning
* Different key letters cause different shift in the plaintext.

he same plaintext letter can produce different
ciphertext depending on the corresponding keyword.

clusion :

The Vigenere cipher is more complex than the
sar cipher because it was uses multiple
ifts controlled by a keyword.

Steganography - hidden information in an image.

-) How can investigation doctect hidden data
within an image?

* steganography is a technique used to hide a confidential information inside an ordinary file, such as an image, audio file or video.

* Metadata analysis : Examine image properties & metadata for unsual information

* File size comparison : Compare the images size with similar original images to identify the unexpected informations

* Statistical analysis : Analysis pixel values & colours patterns to detect the information.

* Steganalysis tools : Use forensis tools to ex the image of concealed information

* File structure examination : Inspect the image file for unexpected data or unusual file structure

(b) Technique : 1 :- Encrypt the confidential data before hiding it

* Encrypt the message using a strong encryption algorithm such as AES.

* Hides the encrypted message inside the image

* Even if someone extract the hidden data they cannot easily understand it without the key

Technique : 2 :

* Use a careful designed steganographic method to embed information in the image.

* Minimise noticeable changes in pixel values

Test the resulting image against statistical steganalysis techniques to reduce the chance of detection.

Conclusion:

Combining encryption with steganography improves confidentially because encryption protects the message's content, while steganography attempts to conceal the existence of the message.

Transposition cipher

(a) Determine the most probable original message:

* The ciphertext can be decrypted using a three rail Rail Fence transposition cipher. In this method, plaintext letters are arranged diagonally across three rails, and the ciphertext is formed by reading the rails row by row.

Step - 1 : Identify the ciphertext

WECRLTEERDSOEEFTEAOCAIVDEN

Step - 2: Reverse the rail arrangement.

* The letters are placed back into their corresponding rail positions, following the zigzag

Step 3 : Read the letters in zigzag order Reading the reconstructed rails diagonally gives :

WEAREDISCOVEREDFLEEATONCE.

With spaces restored, the original message is :

WE ARE DISCOVERED FLEE AT ONCE

(b) Why is this method vulnerable in modern communication systems ?

* No letter substitution : The cipher only rearranges letters ; it does not change their identities.

* Frequency analysis : The frequency of each letter remains unchanged, which can help an attacker analysis the message

* Small key space : A simple three-rail arrangement provides limited security.

* Brute-force attacks : Attackers can test the different arrangements and rail counts to discover the plaintext.

* Lack of modern security : It doesn't provide strong protection against modern cryptanalysis or computer based attacks.

Conclusion: A transposition cipher is unsuitable for protecting sensitive corporate information because its security is limited

c) Propose one practical enhancement :

Enhancement : Use AES encryption.

* The company should replace the traditional transposition cipher with the advanced encryption Standard (AES), a modern symmetric encryption algorithm.

Benefits :

* Provides strong confidentiality when correctly implemented.
* Uses a secret key for encryption and decryption.
* Is designed to resist modern cryptographic attacks.
* Can protect sensitive corporate messages more effectively than a simple classical cipher.

Conclusion :

* Replacing the three-rail transposition ciphes with properly implemented AES encryption would significantly improve message confidentiality.